



**Teacher Copy** — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 8 · STANDARD 6.DS.1

# Statistical Questions and Data

Flagship

Lesson 8-1-flagship · Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## FRONT OFFICE MISSION

### Scout the Hardwood

You are the new data analyst for the Riverside Hoops youth league. The head coach drops a stack of questions on your desk — some will yield rich data full of variety, others just one fixed fact. Master statistical questions and you decide what the front office actually investigates.



## TODAY'S OBJECTIVES

**Content Objective:** I can tell the difference between a statistical question and a non-statistical question.

**Language Objective:** I can explain my reasoning using the words statistical question, data, variability, and survey.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Statistical Question

Data

Variability

Survey

Data distribution



Tap any word to see what it means and a picture.

1

A question that expects a variety of answers and is answered by collecting data is a

2

Facts or numbers collected to answer a question are .

3

How much the values in a data set differ from one another is the

4

A way to collect data by asking many people the same question is a

5

The way data values are spread out across a range is the .

## Reflect — Exit Ticket

Which question is a statistical question? A) How many students are in the school? B) How many hours does each student in 6th grade exercise per week? C) What time does school start? D) How many days are in a week?

A. B) How many hours does each student exercise per week?

B. A) How many students are in the school?

C. C) What time does school start?

D. D) How many days are in a week?

Your answer:

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## Answer Key & Teacher Guide

1. **Notes 1:** Statistical Question
2. **Notes 2:** Data
3. **Notes 3:** Variability
4. **Notes 4:** Survey
5. **Notes 5:** Data distribution
6. **Watch for this mistake:** *A common mistake in Statistical Questions and Data is skipping the key idea: "If you expect the answers to vary, it is a statistical question." — always check your work against this rule before you submit.*
7. **Try It 1:** A. How many rebounds does each player grab per game? — *"How many rebounds does each player grab per game?" is statistical because different players grab different numbers of rebounds — the data anticipates variability. The other three each have one fixed answer.*
8. **Try It 2:** A. Because different players warm up for different amounts of time — *A statistical question anticipates variability. Different players spend different amounts of time warming up, so the collected data will vary from player to player.*
9. **Exit Ticket:** A. B) How many hours does each student exercise per week? — *"How many hours does each student exercise per week?" is statistical because different students exercise different amounts — the data has variability.*